

Vibin Abraham

Theoretical Chemistry and Quantum Computing

Physical & Computational Sciences Division, Pacific Northwest National Laboratory

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Theoretical chemist developing classical and quantum methods for many-body electronic structure, spectroscopy, and quantum simulation. Research spans real-time Green's functions, relativistic GW for quantum materials, tensor-product states, quantum algorithms, and open-system dynamics, with applications to molecular systems and strongly correlated materials. Developing end-to-end computational workflows through collaborations across academia, national laboratories, and industry.

Research Experience

Postdoctoral Research Associate

Jun 2025 – Present

Pacific Northwest National Laboratory, Richland, WA | Advisor: [Bo Peng](#)

- Lead development of fault-tolerant quantum algorithms and efficient classical methods for many-body core-level spectroscopy using real-time Green's functions.
- Develop noise-aware quantum algorithms that learn from noisy real-time dynamics to improve robustness in many-body quantum simulations.
- Build GPU-accelerated real-time coupled-cluster (RT-CC) software to reduce the cost of many-body core-level spectroscopy and enable larger simulations.
- Investigate quantum simulation of strongly correlated materials using downfolded Hamiltonians and effective models, including two-dimensional Hubbard systems.
- **End-to-End Quantum Simulation of HDAC8:** Develop classical-quantum workflows integrating electronic-structure calculations, projector-based embedding, active-space reduction, and quantum simulation for a biologically relevant metalloenzyme. Collaborative project involving IQM, NVIDIA, Argonne National Laboratory, and UCL.
- **Strongly Correlated Materials:** Develop end-to-end quantum simulation workflows for two-dimensional Hubbard systems using downfolded Hamiltonians, connecting electronic-structure modeling with effective many-body representations. Collaborators: Mario Motta, Karol Kowalski, and Bo Peng.
- **Molecular Spin-Qubit Decoherence:** Develop an end-to-end computational workflow for modeling dephasing and open-system dynamics in molecular spin qubits, connecting molecular modeling with quantum dynamical simulations.
- **Quantum Algorithms for Constrained Systems:** Investigate adiabatic gauge potential (AGP)-based optimization for cardinality-constrained systems, including lattice-gas models, to generate improved initial states for quantum calculations.
- **Compact and Noise-Resilient Quantum Methods:** Investigate dual-space formulations for compact generalized eigenvalue problems and noise-learning strategies based on noisy quantum Krylov dynamics.

Postdoctoral Research Associate

Jan 2022 – Jan 2025

University of Michigan, Ann Arbor, MI | Advisor: [Dominika K. Zgid](#)

- Developed and applied relativistic, fully self-consistent *GW* methods for heavy-element molecules and solids, including actinide systems.
- Advanced Green's function electronic structure methods to improve scalability and accuracy for molecular and condensed-phase materials.

Graduate Research Assistant

Aug 2016 – May 2022

Virginia Tech, Blacksburg, VA | Advisor: [Nicholas J. Mayhall](#)

- Designed tensor-product electronic structure frameworks for excitonic processes in extended systems and molecular aggregates.
- Investigated singlet fission and energy transfer in organic photovoltaic materials, with predictions validated by independent experimental studies.
- Co-developed quantum algorithms for excited-state simulations with Los Alamos National Laboratory.

Undergraduate Summer Researcher

May – Jul 2014

Virginia Tech, Blacksburg, VA | Advisor: [T. Daniel Crawford](#)

Investigated the chiral imprint induced by a chiral solute in an achiral solvent.

Education**Ph.D. in Chemistry**

Aug 2016 – May 2022

Virginia Tech, Blacksburg, VA | Advisor: Nicholas J. Mayhall

Dissertation: [Modern Electronic Structure Theory using Tensor Product States](#)**Integrated Master's in Chemistry**

Jul 2011 – Jun 2016

Indian Institute of Technology Bombay, Mumbai | Advisor: Raghavan B. Sunoj

Dissertation: Design of Asymmetric Catalysts using Multivariate Linear Regression

Peer Reviewed Publications

1. J. Adamski, **V. Abraham**, and D. Zgid, All-Electron Relativistic Fully Self-Consistent *GW* Study of Heteronuclear Actinide-Containing Diatomics, *J. Phys. Chem. Lett.* **17**(35), 9997 (2026).
2. **V. Abraham**, P. Senapati, H. Pathak, and B. Peng, Elucidating many-body effects in molecular core spectra through real-time approaches: Efficient classical approximations and a quantum perspective, *J. Chem. Phys.* **164**, 104113 (2026)
3. S. Iskakov, C.-N. Yeh, P. Pokhilko, Y. Yu, L. Zhang, G. Harsha, **V. Abraham**, M. Wen, M. Wang, J. Adamski, *et al.*, Green/weakcoupling: Implementation of fully self-consistent finite-temperature many-body perturbation theory for molecules and solids, *Comput. Phys. Commun.* **306**, 109380 (2025)
4. G. Harsha, **V. Abraham**, M. Wen, and D. Zgid, Quasiparticle and fully self-consistent *GW* methods: an unbiased analysis using Gaussian orbitals, *Phys. Rev. B* **110**, 235146 (2024)
5. **V. Abraham**, G. Harsha, and D. Zgid, Relativistic fully self-consistent *GW* for molecules: Total energies and ionization potentials, *J. Chem. Theory Comput.* **20**, 4579 (2024)
6. G. Harsha, **V. Abraham**, and D. Zgid, Challenges with relativistic *GW* calculations in solids and molecules, *Faraday Discuss.* (2024)
7. M. Wen, **V. Abraham**, G. Harsha, A. Shee, K. B. Whaley, and D. Zgid, Comparing self-consistent *GW* and vertex-corrected G_0W_0 ($G_0W_0\Gamma$) accuracy for molecular ionization potentials, *J. Chem. Theory Comput.* **20**, 3109 (2024)
8. A. Kumar, A. Asthana, **V. Abraham**, T. D. Crawford, N. J. Mayhall, Y. Zhang, L. Cincio, S. Tretiak, and P. A. Dub, Quantum simulation of molecular response properties in the NISQ era, *J. Chem. Theory Comput.* **19**, 9136 (2023)
9. N. M. Braunscheidel*, **V. Abraham***, and N. J. Mayhall, Generalization of the tensor product selected CI method for molecular excited states, *J. Phys. Chem. A* **127**, 8179 (2023)
10. A. Asthana, A. Kumar, **V. Abraham**, H. Grimsley, Y. Zhang, L. Cincio, S. Tretiak, P. A. Dub, S. E. Economou, E. Barnes, *et al.*, Quantum self-consistent equation-of-motion method for computing molecular excitation energies, ionization potentials, and electron affinities on a quantum computer, *Chem. Sci.* **14**, 2405 (2023)
11. **V. Abraham** and N. J. Mayhall, Coupled electron pair-type approximations for tensor product state wave functions, *J. Chem. Theory Comput.* **18**, 4856 (2022)
12. **V. Abraham** and N. J. Mayhall, Revealing the contest between triplet–triplet exchange and triplet–triplet energy transfer coupling in correlated triplet pair states in singlet fission, *J. Phys. Chem. Lett.* **12**, 10505 (2021)
13. **V. Abraham** and N. J. Mayhall, Cluster many-body expansion: A many-body expansion of the electron correlation energy about a cluster mean field reference, *J. Chem. Phys.* **155** (2021) [EDITOR'S PICK]
14. **V. Abraham** and N. J. Mayhall, Selected configuration interaction in a basis of cluster state tensor products, *J. Chem. Theory Comput.* **16**, 6098 (2020)
15. **V. Abraham** and N. J. Mayhall, Simple rule to predict boundedness of multiexciton states in covalently linked singlet-fission dimers, *J. Phys. Chem. Lett.* **8**, 5472 (2017)
16. C. Patel, **V. Abraham**, and R. B. Sunoj, Mechanistic insights and the origin of regioselective borylation in an Iridium-catalyzed alkyl C(sp³)–H bond functionalization, *Organometallics* **36**, 151 (2017)

* Equal contribution.

Preprints

1. **V. Abraham**, B. Jayee, B. Peng, N. P. Bauman, and K. Kowalski, Toward Resilient Many-Body Formulations under Incomplete Correlation Models: A Dual-Space Variational Formulation. [arXiv:2609.04387 \(2026\)](#).
2. **V. Abraham** and B. Peng, Geometry-Dependent Nonlocal Valence Screening Following Core Ionization of the Water Dimer. [arXiv:2608.22268 \(2026\)](#).
3. P. Senapati, **V. Abraham**, Q. Guan, and B. Peng, Unified Uncertainty Quantification Framework Bridging Noisy Quantum Backends Across Variational Quantum Algorithms and Quantum Signal Processing. [arXiv:2607.14392 \(2026\)](#).
4. S. P. Sundar, **V. Abraham**, B. Peng, and A. Asthana, Chemically Decisive Benchmarks on the Path to Quantum Utility. [arXiv:2601.10813 \(2026\)](#).

Manuscript in Preparation

1. **V. Abraham** and B. Peng, Noise-adaptive channel-orbit Krylov subspaces for quantum many-body simulation.

Research Software

pytdcc

Software for real-time coupled-cluster calculations of X-ray spectra.

Green/weakcoupling

Contributed to a framework for simulating realistic materials with Green's function methods.

FermiCG

Contributed to the Julia package for coarse-grained approaches to fermionic many-body problems.

Adapt-VQE

Contributed to simulation code for ADAPT-VQE and quantum self-consistent equation-of-motion (q-sc-EOM) algorithms.

Awards and Achievements

Outstanding Performance Award

Jan 2026

Pacific Northwest National Laboratory

Best Postdoctoral Poster Award

Jun 2022

52nd Midwest Theoretical Chemistry Conference

Graduate Research Award

2021

Department of Chemistry, Virginia Tech

Best Lightning Talk Award

Jul 2020

Virtual Conference in Theoretical Chemistry

IBM Zerner Graduate Student Award

Feb 2020

60th Sanibel Conference on Theoretical Chemistry

JCP Poster Award

Jun 2018

16th International Congress of Quantum Chemistry

INSPIRE Scholarship

Jul 2011 – Jun 2016

Department of Science and Technology, Government of India

IIT-JEE

2011

Ranked in the top 1% among more than 500,000 participants in India.

Teaching Mentoring and Service

Peer Reviewer

Jul 2021 – Present

Physical Review Letters; Physical Review B; Physical Review Research; Journal of Chemical Theory and Computation; Journal of Chemical Physics; Physical Chemistry Chemical Physics.

Graduate and Undergraduate Mentor

2016 – Present

Mentor two Ph.D. interns at PNNL. Previously mentored four graduate and two master's students during doctoral and postdoctoral research, leading to several co-authored publications.

Volunteer Instructor, LCWAA, Mumbai

Aug – Dec 2020

Taught chemistry to high school students from underserved communities during the COVID-19 lockdown; adapted lessons for online delivery and limited resources.

Teaching Assistant, General Chemistry, Virginia Tech

Fall 2016 – Spring 2017

Led undergraduate laboratory instruction in chemical concepts, experimental techniques, data analysis, and problem-solving.

Instructor, CH 105, IIT Bombay

Summer 2015

Taught introductory organic chemistry to 100 students as part of a four-member instructional team.

Volunteer Instructor, Abhyasika, Mumbai

Aug 2012 – Aug 2016

Taught mathematics and English grammar to students from low-income communities.

Invited Talks

GW Method for Molecular Systems

May 2026

IITM Workshop on Theoretical and Computational Chemistry

V. Abraham

Understanding Many-Body Effects in Core Level Spectroscopy

Apr 2026

University of North Dakota, Grand Forks

V. Abraham and B. Peng

Conference Presentations

Elucidating Many-Body Effects in Molecular Core Spectra through Real-Time Approaches

2026

APS March Meeting, Denver

V. Abraham, P. Senapati, H. Pathak, and B. Peng

Quantum and classical methods for strongly correlated systems

2025

Future of Chemistry Symposium, TIFR, Mumbai

V. Abraham

Estimating the accuracy of pseudopotential based *GW* method at different levels of self consistency using Gaussian orbitals

2023

APS March Meeting, Las Vegas, Nevada

V. Abraham, M. Wen, G. Harsha, and D. Zgid

Modern electronic structure theory using tensor product states

Spring 2021

ACS National Meeting, Atlanta, Georgia

V. Abraham and N. J. Mayhall

Tensor Product Selected CI: Compact wave function for strongly correlated cluster-able systems

Jun 2020

Virtual Conference of Theoretical Chemistry; Graduate Student Honorable Mention

V. Abraham and N. J. Mayhall